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Fraud Detection on Medicare Claims Using SQL-Based Feature Engineering and Machine Learning

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ABSTRACT Healthcare fraud in Medicare always results in substantial financial losses and is becoming harder to detect as claims data become increasingly sophisticated and imbalanced. In this paper, we present a complete pipeline combining an SQL-based feature store, supervised binary classification, and unsupervised clustering to detect and categorize fraudulent activity. Unsupervised clustering is applied only to ground-level fraudulent cases. Then we classified the fraudulent providers and clustered the high-risk providers into separate fraud type groups. Using inpatient, outpatient, and beneficiary tables from a Medicare dataset, we engineered reusable provider- and claim-level features for model training and evaluation. This included rule-based indicators for duplicate claims, upcoding, and overcharging, as well as temporal aggregates and peer-normalized z-scores. We evaluated multiple supervised learning models to classify fraudulent providers; a regularized logistic regression model achieved the strongest performance with an AUPRC of 0.73 and an ROC-AUC of 0.94. Furthermore, we applied the HDBSCAN algorithm, Principal Component Analysis (PCA), and association rule mining to known fraudulent cases and high-risk providers. This unsupervised approach allowed us to cluster and characterize distinct fraud typologies successfully. In general, this integrated methodology establishes a robust evaluation benchmark and demonstrates an effective, scalable approach to profiling complex fraud schemes in healthcare.

INDEX TERMS Association rule mining, HDBSCAN fraud clustering, Healthcare insurance fraud types, Medicare fraud detection, SQL-feature engineering, Tree-based ensemble learning, Unsupervised anomaly detection.

I. INTRODUCTION

HEALTHCARE fraud and abuse divert limited resources from patient care and impose substantial financial losses on a large public insurance program such as Medicare. It is estimated that 3-10% of total healthcare expenses are subject to fraud and waste each year, amounting to tens to hundreds of billions of dollars globally and, on average, 300 billion USD annually in the US [1]- [3]. This waste also eroded trust in health systems [3], [4]. Reviews show that as health insurance costs rise in many countries, fraudulent activities are also increasing proportionally, compromising patient safety and raising questions about the quality of care [2], [4]. Medicare claims provide a lot of information regarding patients, providers, diagnoses, procedures, deductibles, and reimbursements for detection. However, these features are highly imbalanced and multidimensional [1], [2], [5]. With the multidimensional dataset, detection is sophisticated, as fraud cases are rare among all claims [1]- [3], [6]- [10]. Rule-based systems are easy to implement but offer less scope for automation in fraud detection and can lead to high false-positive rates [2]- [4]. These limitations have inspired comprehensive research in healthcare to invent effective strategies to prevent fraud [1]- [4], [11], [12].

The primary objective of this study is to develop and evaluate a comprehensive, hybrid pipeline for detecting and characterizing Medicare fraud. By integrating an SQL-based feature store with supervised classical machine learning models and unsupervised clustering algorithms (HDBSCAN), this research aims to establish a scalable benchmark to overcome the limitations of traditional rule-based systems and highly imbalanced claims data. To achieve our objective, this research introduces the following contributions to the field of healthcare fraud detection:

- Scalable SQL-Based Feature Engineering: Instead of relying on memory-bound Python data frames (e.g., pandas), we develop a domain-specific feature store natively in PostgreSQL. For complex computational workloads, rolling temporal aggregates and peer-normalized indicators for upcoding and overcharging at the database layer provide a scalable and reproducible framework that effectively resolves the class imbalance in Medicare data before model ingestion.
- Classical Machine Learning Benchmark: We show that when equipped with robust feature engineering, models like logistic regression can match or exceed the predictive power of tree-based ensembles, achieving a ROC-AUC of 0.94 and AUPRC of 0.73.
- Unsupervised Fraud Typology Mapping: Moving beyond binary classification, we introduce a methodology combining HDBSCAN clustering and association rule mining to successfully isolate and label real-world fraud schemes such as high-revenue inpatient overcharging (Cluster 1) versus high-volume outpatient overbilling (Cluster 0 and 2).

II. RELATED WORK

Recent studies present a variety of audit-based checks and ML techniques for fraud detection in health insurance, including statistical models, ensemble learning, support vector machines, deep learning, and hybrid pipelines [1]- [4]. Du Preez et al. did not identify a single best approach in their review of machine learning methods [1]. The main reasons are differences in dataset features, different types of fraud, and evaluation techniques. Having access to the ground truth is also a challenge here. Kumaraswamy et al. worked on data-mining methods for fraud detection [2]. They focused primarily on the significant financial loss, evolution of fraud schemes, and differences in evaluation between academic models and real-world systems. Thaifur et al. reported the discovery of various types of fraudulent activities and focused only on providers [4]; they skipped inspections at the claim level. Razzaq and Shah highlighted challenges related to imbalanced data, scarcity of trustworthy labels, privacy concerns, and the practicality of the models in terms of usability [3].

Supervised models were used to classify fraudulent claims and providers. Nabrawi and Alanazi evaluated several supervised models on Saudi insurance claims, and their work showed that tree-based models and the SVM model performed well when the ground truth for fraud cases was available [13]. Hamid et al. studied the CMS DE-SynPUF claims and found that gradient boosting performed well despite class imbalance and that provider-level features were carefully implemented [11]. Liu and Vasarhelyi proposed a clustering model using geographic and payment data [14]. Their approach combined claim and provider features to flag suspicious cases [14]. Kumar and Sountharrajan built an optimized deep learning model with proper hyperparameter tuning for fraud detection and noted challenges with feature engineering and class imbalance [15]. Similarly, F. Ahammed et al. developed a deep neural network framework that integrated feature selection (Chi-squared method) and data sampling techniques (SMOTE) to address the challenges of high dimensionality and class imbalance in Medicare claims data [16]. Yet their model struggles to meet the computational resources required for real-time detection on live-streaming datasets and relies on provider-level feature engineering [10]. Hancock et al. proposed explainable ensemble models for Medicare data using gradient-boosted trees; however, their reliance on tree depth and feature reduction lacks the transparency of explicit rule-based indicators [17].

Unsupervised anomaly detection methods are found to be more suitable when working with limited labels for fraud cases. Bairy et al. tested Isolation Forest, Local Outlier Factor, KNN, and autoencoders on a healthcare dataset; their methods can spot rare anomalies that need further review [5]. De Meulemeester used deep embeddings to flag anomalies and added necessary explanations for the investigators [18]. Kanksha et al. flagged fraud cases without having enough fraud labels [19]. Settipalli and Gangadharan worked on WMTDBC, which converted correlated claim data into independent variables. They also used distance-based detection, which improved the detection of multivariate extremities [20]. Massi et al. first grouped providers into peer clusters and then identified each outlier within each cluster [12]. Saldamli et al. worked on blockchain systems to enhance transaction security and decentralize claims data [21]. Matloob et al. proposed a two-module framework utilizing sequence analysis, a rule engine, and a prediction-based engine to identify healthcare insurance fraud [22]. Their initial data validation was hindered by the difficulty of acquiring confidential employee insurance data.

Although supervised models achieve high predictive performance [11], [13], [15], they struggle in the absence of ground-truth labels and often lack claim-level granularity [4]. Furthermore, benchmark ensemble methods, which are highly accurate, often sacrifice the clinical transparency required for auditing (Hancock et al. [17]). In contrast, unsupervised methods [5], [12], [19], [20] successfully bypass the need for labels but flag broad anomalies that overwhelm investigators and lack predictive probability scoring. With scalable SQL-based feature engineering, supervised provider classification, and unsupervised clustering, our approach mitigates class imbalance and produces interpretable, actionable fraud typologies. All of these works are briefly summarized in Table 1.

TABLE 1: Summary of State-of-the-Art Methodologies and Research Gaps

References	Methodology	Strengths	Research Gaps
Thaifur et al. [4]	Fraud Discovery	Successfully categorizes different types of fraudulent activities	Focuses exclusively on the provider level, completely skipping critical inspections at the individual claim level
Bairy et al. [5] & Kanksha et al. [19]	Unsupervised Models: Isolation Forest and Local Outlier Factor (LOF) algorithms	Best performance in Isolation Forest & Handling Complex Data	Need for Advanced Architectures, Limited to a Single Dataset
F. Ahammed et al. [10]	Chi-squared Method & SMOTE	Reduces model complexity and overfitting, highly effective at minimizing false negatives	Model struggles with the computational resources required for real-time detection on live-streaming datasets
Hamid et al. [11]	Association Rule Mining & Unsupervised Models	Fewer False Positives, Adaptable & Robust System	Lack of representation of the diversity and complexity of real-world fraud scenarios
Settipalli & Gangadharan [20] & Massi et al. [12]	WMTDBC & Peer Clustering	Successfully isolates outliers within peer groups (e.g., multivariate extremities)	Distance and peer-based models identify extremes but lack a supervised scoring mechanism to assign definitive fraud probabilities
Nabrawi & Alanazi [13]	Supervised Models (Tree-based & SVM)	Achieves strong classification performance on Saudi insurance claims	Small sample size and a limited number of features, which restricted its ability to cover all aspects of fraud
Liu & Vasarhelyi [14]	Geographic & Payment Clustering	Spots concurrent provider fraud, subscriber fraud & collusive fraud	Flags "suspicious" claims only for further investigation and cannot definitively prove a claim is fraudulent on its own
Kumar & Sountharrajan [15]	Novel Deep Learning Framework: EHOA-CNN-12	Balanced Architecture, Improved Optimization & High Accuracy	High Computational Requirements, Lack of Interpretability, Lack of Adversarial Testing
Hancock et al. [17]	Ensemble Supervised Feature Selection Technique	Massive Dimensionality Reduction, Computational Efficiency	Lack of Domain Diversity, Reliance on Retrospective Labels
De Meulemeester [18]	Categorical Embeddings	Utilizes embeddings to discover unknown fraud & highly interpretable	Conducted purely for research purposes on historical data & has not yet been implemented for active, operational goals
Saldamli et al. [21]	Blockchain Systems	Provides advanced security and decentralization for transactions	Focuses on data security infrastructure rather than the algorithmic detection of sophisticated billing anomalies
Matloob et al. [22]	Sequence Analysis, Rule Engine & Prediction-Based Engine	Reached an average accuracy of 85% and analyzed complex patient sequences across more than 62 specialties	Future research should test the framework on much larger datasets to better evaluate the model's competence

III. METHODOLOGY

The complete sequence of operations for our hybrid approach from SQL-based data curation to unsupervised fraud typology mapping is formally outlined in Algorithm 1.

A. DATA SOURCES AND STUDY DESIGN

We used the Medicare dataset from Zenodo, which comes with 4 training files and 4 test files. Both have provider, inpatient, and outpatient claims, as well as beneficiary tables. The training provider table includes a *PotentialFraud* flag. Inpatient and outpatient tables contain rows with claim IDs, beneficiary IDs, provider IDs, dates, reimbursed amounts, deductibles, and diagnosis/procedure codes. The inpatient table also has admission and discharge dates and diagnostic-related fields. The beneficiary table contains basic patient information, demographics, and the patient's chronic health condition. We computed mean, median, and z-scores on payments and length of stay. We also did the same for total claims, reimbursement amounts, and unique beneficiaries over 30/90/365 days to classify fraudulent providers. We utilized claims data for feature construction and performed unsupervised clustering on fraud-only claims. For supervised learning, we trained our models on 70% of the data and held out 30% for evaluation. We then applied the trained models to unlabeled test data to examine risk scores and labels to characterize the provider clusters.

B. DATA CLEANING AND SQL FEATURE STORE

The decision to implement feature engineering in SQL provides a significant architectural advantage over existing methodologies. Medicare claims are massive relational datasets that include inpatient, outpatient, and beneficiary tables. Traditional approaches often extract raw data into memory-bound environments for processing, introducing scalability bottlenecks. Using PostgreSQL to build a layered feature store, we push the complex computational workload directly to the database level. All data processing was implemented in PostgreSQL using a layered design. We loaded raw CSV files into a staging schema with all fields as text and basic checks on structure and column counts. The cleaned data was moved to a curated schema, where the placeholder strings for missing values were converted to NULL and the columns were cast to the appropriate types. We went through data processing steps to verify unique provider and beneficiary IDs. We ensured that all features had non-null values across the inpatient and outpatient tables. We also looked into valid foreign keys and non-negative payments, length of stay, and other numeric features. After handling those checks, we put together a mart schema with a fact table and feature tables focused on claims and providers. These tables contain details on length of stay, code counts for diagnoses and procedures, duplicates, possible overcharges, peer-normalized z-scores, and 30-, 90-, and 365-day aggregates.

Algorithm 1 Hybrid Healthcare Fraud Detection and Characterization Pipeline

Require: Raw Medicare Data $\mathcal{D}_{raw} = \{\text{Claims}_{IP}, \text{Claims}_{OP}, \text{Beneficiaries}, \text{Providers}\}$
- Ground truth label set Y where $y_i \in \{0, 1\}$ (1 = Fraudulent Provider)

Ensure: Trained classification model M_{best} and optimal threshold τ_{opt}
- Provider fraud risk scores S_{risk}
- Identified claim clusters $C^{IP}_{claims}, C^{OP}_{claims}$ and provider clusters $C_{providers}$
- Association rules R_{assoc} for fraud typologies

Phase 1: SQL-Based Feature Engineering & Data Curating
Clean and standardize $\mathcal{D}_{raw}$ into unified fact table F
for each provider i in F **do**
  Compute rolling aggregates (30, 90, 365 days) for claim counts, patient volumes, and reimbursements
end for
for each claim c in F **do**
  Compute peer-normalized z-scores (μ, σ):
  if IP **then**
    Group by DRG $\rightarrow Z_{reimb} = (X_c - \mu_{DRG})/\sigma_{DRG}$
  else if OP **then**
    Group by Diagnosis Prefix $\rightarrow Z_{reimb} = (X_c - \mu_{Prefix})/\sigma_{Prefix}$
  end if
  Apply binary rule-based indicators I:
  $I_{duplicate} \leftarrow 1$ **if** exact match **OR** (date ± 1 day **AND** amount $\pm 1\%$)
  $I_{upcode} \leftarrow 1$ **if** $(Z_{reimb} > 2.5)$ **AND** $(Z_{LOS} < -0.5$ **OR** $LOS < 2$ days)
  $I_{overcharge} \leftarrow 1$ **if** $(Z_{reimb} > 3.0$ **OR** $X_{reimb} > Q3 + 1.5 \times IQR)$
end for
Aggregate claim-level features and I flags to generate provider-level feature matrix X_{prov}
Phase 2: Supervised Provider Fraud Classification
Join X_{prov} with labels Y
Partition data: $X_{train}, X_{test} \leftarrow \text{StratifiedSplit}(X_{prov}, 70/30)$
Apply Standard Normalization: $X_{scaled} \leftarrow (X - \mu_{train})/\sigma_{train}$
Apply class-weighting to mitigate label imbalance
Train classifiers $\in$ {Logistic Regression (Elastic-Net), Random Forest, Extra Trees, LightGBM, XGBoost, CatBoost}
Evaluate models using 5-fold Cross Validation optimized for AUPRC
Select $M_{best} \leftarrow$ Logistic Regression
Define $\tau_{opt} \leftarrow$ Threshold maximizing F1-Score on PR Curve
Generate probability scores: $S_{risk} \leftarrow M_{best}.\text{predict\_proba}(X_{scaled})$
Phase 3: Unsupervised Fraud Typology Mapping
Claim-Level Analysis:
Filter X_{claims} where $Y = 1$ (Fraud-only cases)
Apply HDBSCAN distinctly to Inpatient and Outpatient subsets $\rightarrow C^{IP}_{claims}, C^{OP}_{claims}$
Run Apriori algorithm within clusters to extract diagnosis-procedure association rules R_{assoc}
Provider-Level Analysis:
Filter X_{prov} where $S_{risk} \geq \tau_{high\_risk}$ **OR** $Y = 1$
Apply HDBSCAN (*min_cluster_size*=10) to isolate high-risk provider groups $C_{providers}$
Project $C_{providers}$ into 2D space using Principal Component Analysis (PCA)
return $M_{best}, \tau_{opt}, S_{risk}, C^{IP}_{claims}, C^{OP}_{claims}, C_{providers}, R_{assoc}$

C. FEATURE ENGINEERING AND RULE-BASED INDICATORS

We implemented simple rules on both claim and provider features to capture duplicate billing, upcoding, overcharging, and other fraud patterns. We worked on the amount of reimbursement, the deductible, the length of stay, and the counts of diagnoses and procedures, depending on the type of patient. We also detected exact or near duplicates by comparing the same patient and provider in a short period of time. We then calculated peer-normalized z-scores and IQR thresholds for reimbursement payments and grouped patient records. To construct peer-normalized features, claims were grouped by their respective clinical categories (Diagnosis Related Group for inpatient claims and Primary Diagnosis

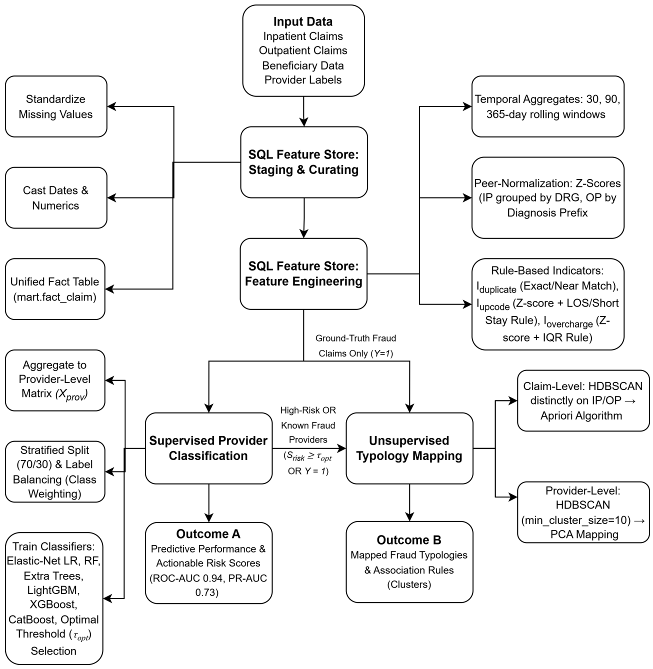


FIGURE 1: Experimental design diagram of the proposed methodology. The pipeline shows the input of Medicare data, the SQL-based feature engineering process, and the evaluation using supervised machine learning models and unsupervised HDBSCAN clustering.

Prefix for outpatient claims) across 30-, 90-, and 365-day windows. The standardized deviation Z for the reimbursement amount of a given claim c was mathematically formulated as:

$$Z_c = \frac{X_c - \mu_p}{\sigma_p} \tag{1}$$

where X_c represents the claim's reimbursement amount, while μ_p and σ_p represent the mean and standard deviation of all claims within the same peer group p. X_c values were winsorized at the 99th percentile before calculation to prevent extreme outliers from skewing the peer mean [17]. We then observed duplicates, upcoding, and overcharging to flag suspicious claims and used these inputs to train the models in the next steps.

D. EXPLORATORY ANALYSIS AND FEATURE SELECTION

After the initial setup of the SQL feature store was complete, we proceeded with the exploratory analysis and feature selection in Python. Using SQLAlchemy and pandas, we loaded the claim- and provider-level feature tables from the database to inspect the basic distribution of the data. To evaluate skewness and long tails in reimbursement amounts, deductibles, and inpatient length of stay, we plotted histograms and boxplots, which confirmed expected tail behavior. We used IQR- and median-based z-scores to flag extreme outlier claims, then compared them with overcharge indicators and inpatient claims. A claim's reimbursement Xc was flagged as a hard overcharge outlier if it satisfied the following condition:

$$X_c > Q_3 + 1.5 \times (Q_3 - Q_1) \tag{2}$$

where Q_1 and Q_3 denote the 25th and 75th percentiles of the reimbursement amounts within the claim's respective peer group. We used pairwise correlation coefficients and inflation factors to remove highly collinear features, as those are redundant in the feature space. We were interested in the shared ground of recent reimbursement and claim volumes. Therefore, the analysis enabled us to work on a meaningful set of provider features.

E. UNSUPERVISED FRAUD-ONLY CLAIM CLUSTERING

We clustered fraud-only claims to understand the diversity in fraudulent behaviors. We first distinguished fraudulent providers using training labels and grouped all associated claims by type of inpatient and outpatient. We stored those fraudulent records along with the extracted features in separate tables for inpatient and outpatient claims. We identified missing values and replaced them with the median in all records. We scaled features to ensure their contributions to the output were equal, which also sped up the optimization. It also prevented bias toward larger numeric features. We then applied the HDBSCAN clustering algorithm in a clear way to inpatient and outpatient claims [23]- [27]. Our goal was to find clusters of different sizes and densities, and claims that fell under the minimum number of clusters would be considered noise [25]- [27]. We made sure the minimum sizes of clusters and samples were tuned for better accuracy. We also visualized the structure via PCA projections using different colors on the clusters. After that, we ran association rule mining within those clusters to characterize unique types of fraudulent claims.

To extract meaningful diagnosis-procedure links, the Apriori algorithm [11], [28]- [30] evaluates rules of the form $A \rightarrow B$. The strength and reliability of these rules were measured mathematically using Support, Confidence, and Lift, defined as:

$$\text{Support}(A \rightarrow B) = P(A \cap B) \tag{3}$$

$$\text{Confidence}(A \rightarrow B) = \frac{P(A \cap B)}{P(A)} \tag{4}$$

$$\text{Lift}(A \rightarrow B) = \frac{P(A \cap B)}{P(A) \times P(B)} \tag{5}$$

Rules were filtered according to a minimum Support of 0.02, a Confidence of 0.60, and a Lift threshold *(Lift > 1.2)* to statistically ensure that the fraudulent links identified occurred more frequently than expected under random independence.

F. SUPERVISED PROVIDER FRAUD CLASSIFICATION

We applied supervised learning to classify providers as fraudulent or non-fraudulent based on the collected claim features. We trained the models only on labeled training providers, as we know these are absolutely fraudulent cases. We split the fraud cases from the training dataset into a 70/30 holdout set, using 70% for training and 30% for evaluation. This strategy preserved the proportion of fraudulent to non-fraudulent claims. We compared performance using multiple classifiers on the same set of features. We used a linear model (logistic regression with elastic-net regularization) and tree-based models (Random Forests, Extra Trees, LightGBM, CatBoost, XGBoost) for the benchmark evaluation of precision, recall, F1-score, AUPRC, and ROC-AUC [16]. Based on those models, we used class weighting or imbalance-aware parameters to address the scarcity of fraudulent providers. Our optimal-performing model was Regularized Logistic Regression, and it is important to define its objective function. To handle the severe class imbalance problem, we introduced a heuristic class weight (ω_i) into the cross-entropy loss function [16], [31]- [34]. Furthermore, we applied an Elastic-Net penalty to prevent overfitting across the high-dimensional feature space. The model optimized the coefficients (β) by minimizing the following loss function $L(\beta)$:

$$L(\beta) = -\frac{1}{N}\sum_{i=1}^{N} \omega_i \left[y_i \log(\widehat{y}_i) + (1-y_i)\log(1-\widehat{y}_i)\right] + \lambda\left(\alpha\|\beta\|_1 + \frac{1-\alpha}{2}\|\beta\|_2^2\right) \tag{6}$$

where N is the number of providers, y_i is the true label, $\hat{y}_i$ is the predicted probability λ controls the overall regularization strength, and α determines the mix between L_1 (Lasso) and L_2 (Ridge) penalties. We tuned the hyperparameters using randomized search with 5-fold cross-validation and then optimized the AUPRC curve. We did not use the fixed 0.5 cutoff threshold, as it won't always yield the best result in this scenario. We chose the probability threshold that gave us the maximum F1-score on the AUPRC curve. After completing the performance tests, we found logistic regression to be the most performant model, and these performance gains have mostly been achieved by the engineered features rather than the model complexity.

G. PROVIDER FRAUD-TYPE CLUSTERING

To understand the diversity in fraud patterns, we went beyond classifying a provider as fraudulent or not; we sought to characterize fraud activities within the high-risk provider group [29]. We characterized providers already labeled as fraudulent and test providers whose fraud scores have already passed the high threshold. We pulled out the same standardized features that were used in supervised models. These features were reimbursement amounts, claim volumes over an aggregated timeframe, the count of distinct patients, peer-normalized inpatient and outpatient z-scores, and rule scores for duplicate, upcoding, and overcharging claims. We then applied the HDBSCAN algorithm with a minimum cluster size of 10 and a small minimum sample value. This formed clusters with varying sizes and densities, and some providers were scattered throughout the plot and were considered noise. We visualized the structure using principal components and colored points by cluster label, with noise in gray. For each cluster, we produced a labeled summary against the high-scoring mix and interpreted these fraud types as high-revenue inpatient overchargers, high-volume outpatient overbillers, and others.

IV. RESULTS

A. DATA CHARACTERISTICS AND EXPLORATORY ANALYSIS

Training data includes 40,474 inpatient claims, 517,737 outpatient claims, 138,556 distinct patients, and 5410 labeled providers, along with ground-level information on fraud or non-fraud. We found reimbursed amounts highly skewed, as expected in Medicare billing. In most cases, the claims had moderate payments, and a small fraction of the claims were very costly, which produced a long right tail (Figure 2). There were also variations in the inpatient and outpatient claims. Inpatient claims had a higher median reimbursement amount and a wider spread, confirming that inpatient services are more expensive and variable. In contrast, outpatient services show the claims count by length of stay, diagnosis vs. procedure counts, and reimbursement vs. coding intensity (Figure 3). We flagged high-cost claims using interquartile range and median absolute deviation rules. We kept a small set of potential outliers as possible fraud and unusual cases. After preprocessing the data, we found no missing values in the key features. Correlations and variance inflation factors (Figure 4) produced a compact feature set for unsupervised and supervised analyses.

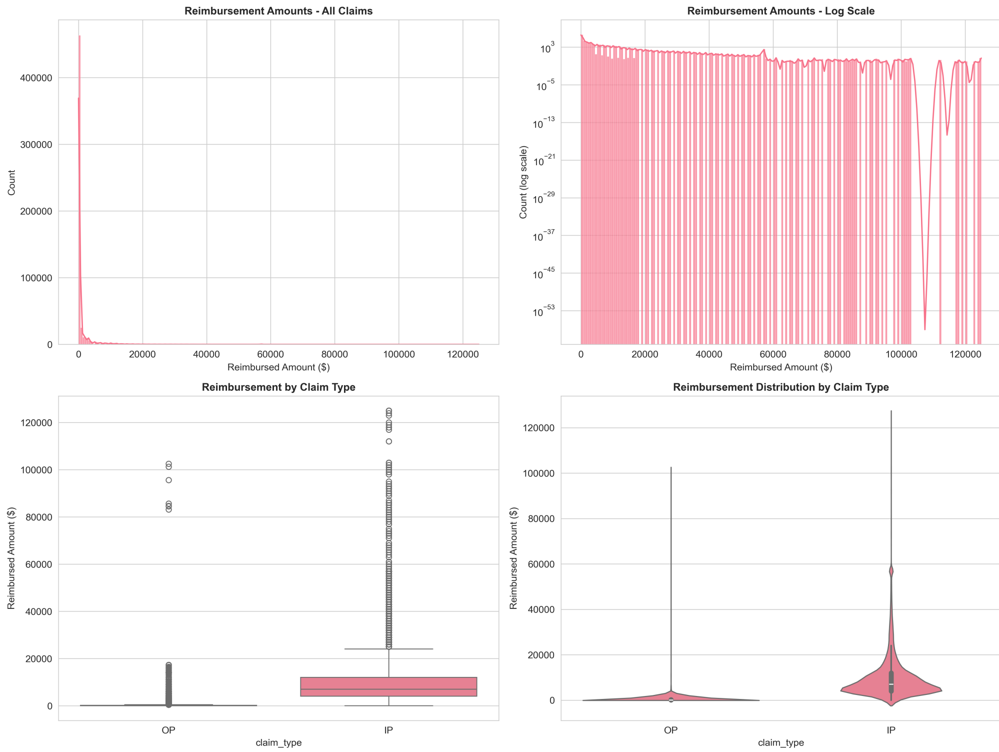


FIGURE 2: Histogram and density plots of reimbursed amounts for inpatient and outpatient claims, showing a heavy right skew and a small number of very high-cost claims.

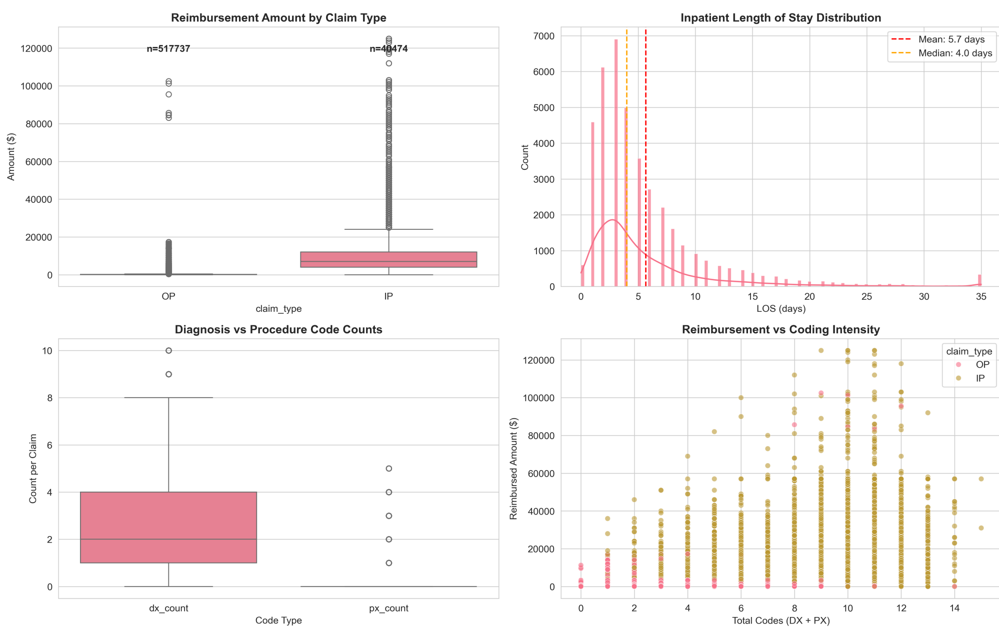


FIGURE 3: Boxplots of reimbursed amounts and inpatient length of stay, highlighting distributional differences between claim types.

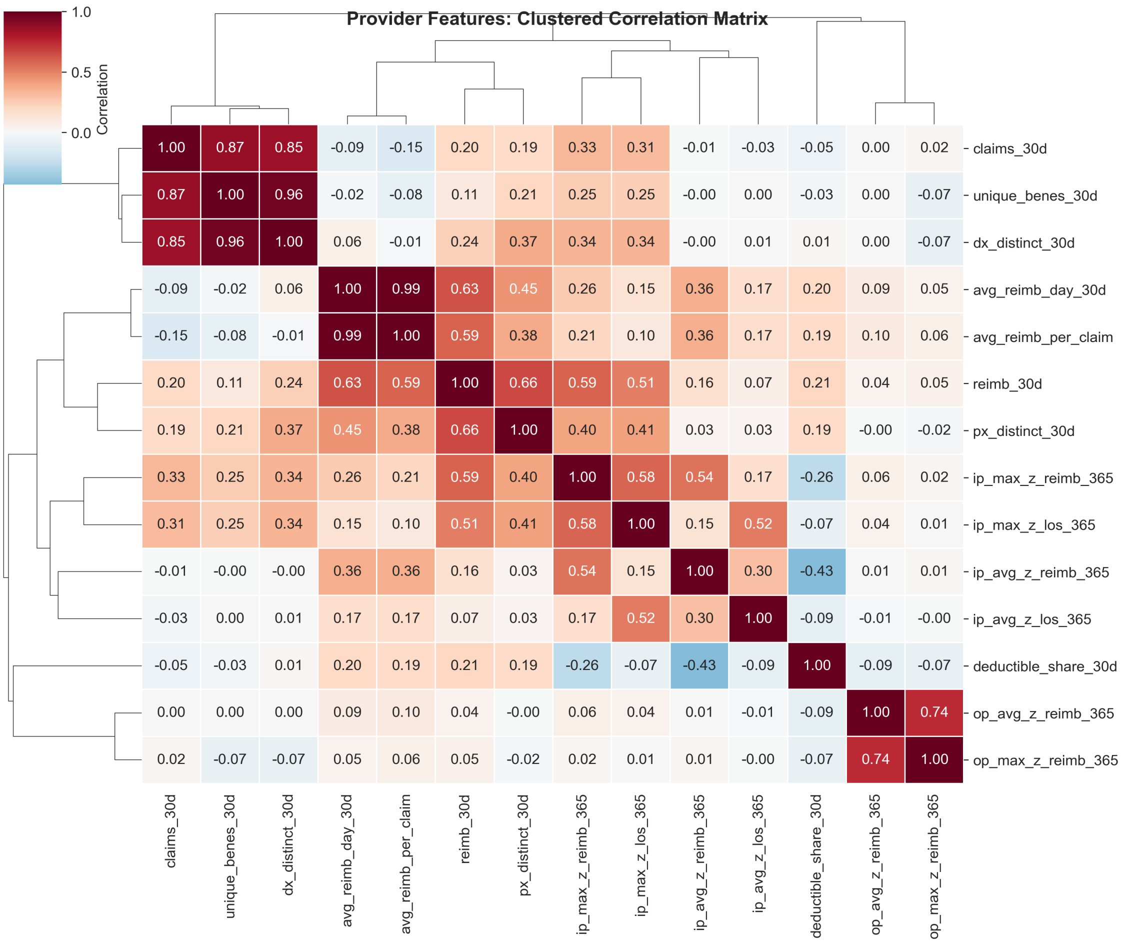


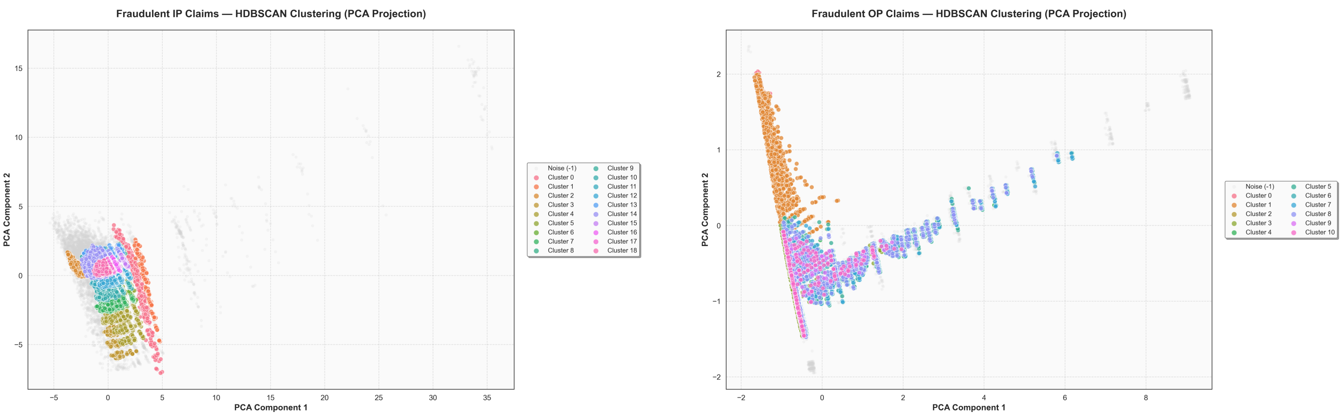
FIGURE 4: Clustered Pearson correlation heatmap of provider-level features, used to identify highly collinear variables and guide feature selection.

B. UNSUPERVISED CLAIM-LEVEL PATTERNS AMONG FRAUDULENT CLAIMS

To understand the diversity of fraudulent behavior, we focused only on fraud-only cases. We identified 22,640 fraudulent inpatient and 8,408 fraudulent outpatient claims out of 558,211 claims. After standardizing the reimbursement amounts, code counts, rule flags, and peer-normalized scores, we implemented HDBSCAN on inpatient and outpatient records. HDBSCAN produced several clusters, and claims that did not belong to any cluster were considered noise. Figure 5(a) showed inpatient fraudulent clusters of high payments and long stays, as we obtained high reimbursement/length-of-stay z-scores. Figure 5(b) showed the visualizations of high-payment groups of fraudulent outpatient claims. Most of the outpatient clusters had low- to moderate-paid claims with very few higher-paid claims. Association rule mining within clusters (Table 2) showed small diagnosis-code sets in outpatient clusters and diagnosis-procedure pairs in high-payment inpatient clusters. The higher lift metrics measure the coordination of these schemes. For example, OP Cluster 1 shows rules with Lift values exceeding 10, indicating that these diagnostic combinations happen over 10x more frequently in fraudulent claims than by random chance. This suggests systematic intentional misrepresentation of related diagnoses rather than isolated clerical errors.

C. PROVIDER-LEVEL FRAUD CLASSIFICATION PERFORMANCE

We evaluated supervised models for classifying fraudulent providers across all cases. On the 30% hold-out evaluation trainset, all models achieved ROC-AUC around 0.93-0.94, indicating that those models were good at telling apart fraudulent and non-fraudulent providers. All the performance metrics are reported in Table 3. Logistic regression with elastic-net regularization resulted in the best AUPRC of around 0.73 and ROC-AUC of around 0.94. After threshold tuning, we got a precision of around 0.59, a recall of around 0.74, and an F1-score of around 0.66. Tree-based models (Random Forests, LightGBM, XGBoost, Extra Trees, CatBoost) [16] achieved similar ROC-AUC but weaker AUPRC and could not surpass the logistic regression model (Figure 6). The linear model with its AUPRC of 0.73 carries important practical



(a) Fraudulent Inpatient Claims (b) Fraudulent Outpatient Claims

FIGURE 5: Two-dimensional PCA projections of standardized fraud-only claims. (a) HDBSCAN clusters of inpatient claims characterized by high payments and extended stays. (b) Outpatient clusters highlighting high-payment groups of fraudulent outpatient claims. Points are colored by cluster labels, with noise shown in grey.

TABLE 2: Association rules for diagnosis and procedure code patterns within selected inpatient and outpatient HDBSCAN claim clusters. Results include antecedent code sets, consequent codes, support, confidence, and lift values for fraud-only clusters.

Claim Type	Cluster	Antecedents Str	Consequents Str	Support	Confidence	Lift
OP	1	280+285	588	0.039	0.766	10.127
OP	1	285+588	280	0.039	0.688	9.462
IP	18	995	38	0.031	0.645	8.896
OP	1	V56	588	0.024	0.637	8.425
OP	1	588	280	0.046	0.612	8.412
OP	1	280	588	0.046	0.636	8.412
IP	14	8154	715	0.031	0.863	7.415
IP	13	7935	820	0.051	0.865	7.398

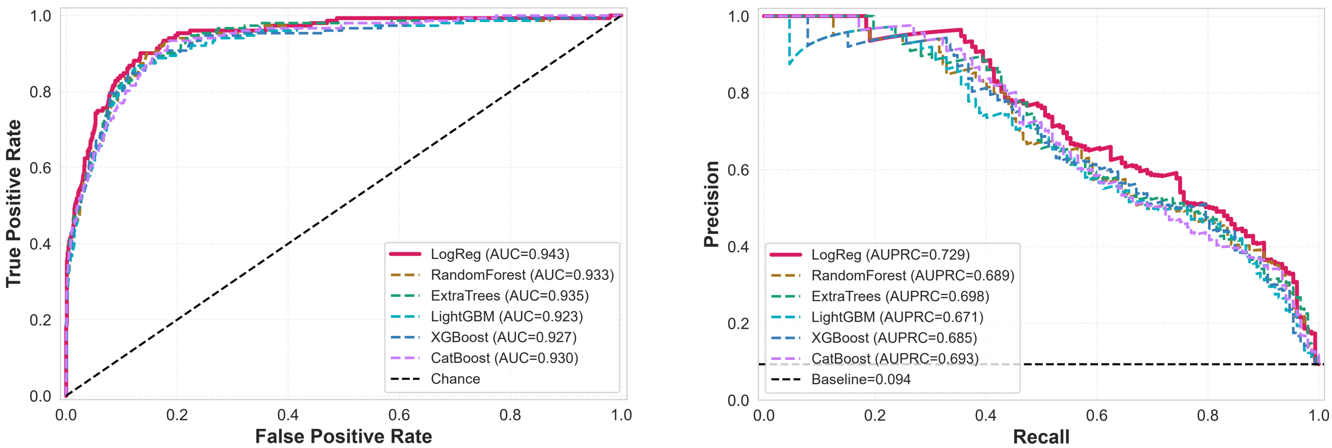
implications. In heavily imbalanced datasets, AUPRC is much more sensitive to false positives than ROC-AUC. The success of the regularized logistic regression indicates that the peer-normalized Z-scores and rolling temporal aggregates established clear linear separability between normal and fraudulent billing behaviors. Therefore, the pipeline effectively discards the need for computationally heavy tree ensembles. Fraud risk scores from the linear model placed most non-fraudulent providers near 0 and fraudulent providers at higher values (Figure 7). Big reimbursements, claim values, z-scores in peer standard feature sets, and rule indicators of duplicates and overcharges contributed the most to these high scores. We then applied those scores to the test providers to define the high-risk providers for clustering in Section IV-D.

D. PROVIDER FRAUD-TYPE CLUSTERS

Among the high-risk class of 570 providers, HDBSCAN identified 4 clusters and labeled 334 providers as noise, meaning that many high-risk providers were not assigned to any cluster. Figure 8 shows compact regions for clusters with mixed noise. The largest cluster (Cluster 3) comprises 180 provider points, characterized by high z-scores for reimbursements and length of stay, but moderate counts of duplicates and overcharges. Cluster 1 (36 providers) comprises high-revenue providers with high inpatient z-scores, potentially indicating upcoding. Cluster 0 and Cluster 2 (10 providers each) account for the high number of outpatient billers with duplicates and overcharge flags. Table 4 shows the results of the fraudulent clusters. This behavioral segmentation provides massive utility for real-world investigations. Instead of flagging a provider with a binary ‘suspicious’ label, this unsupervised HDBSCAN technique and PCA mapping allow auditing agencies to route specific providers to specialized investigative teams. For example, investigators can easily differentiate between a scheme based on high-volume outpatient duplication (Cluster 0) versus severe inpatient upcoding (Cluster 1), allowing for targeted intervention strategies.

TABLE 3: Classification performance of supervised models on the provider hold-out set. Results are reported for logistic regression (elastic net), Random Forests, ExtraTrees, LightGBM, XGBoost, and CatBoost in terms of accuracy, precision, recall, F1 score, AUPRC, and ROC-AUC.

Model	Accuracy	Precision	Recall	F1-Score	AUPRC	ROC-AUC
LogReg	0.93	0.59	0.74	0.66	0.73	0.94
ExtraTrees	0.91	0.53	0.74	0.61	0.70	0.93
CatBoost	0.92	0.57	0.64	0.60	0.69	0.93
RandomForest	0.93	0.62	0.61	0.61	0.69	0.93
LightGBM	0.91	0.51	0.78	0.61	0.68	0.93
XGBoost	0.91	0.51	0.78	0.61	0.68	0.93



(a) Receiver Operating Characteristic (ROC)
(b) Precision-Recall Curve (PRC)

FIGURE 6: Performance curves for supervised provider classification. (a) ROC curves demonstrating high separability across models. (b) Precision-Recall curves highlighting the superiority of the regularized Logistic Regression model on the heavily imbalanced minority class.

TABLE 4: Provider fraud-type clusters identified by HDBSCAN, showing the number of providers, labeled fraud cases in training, high-scored test providers, and median annual reimbursement and claim volumes.

cluster	n_providers	n_train_fraud	median_reimb_365d	median_claims_365d
0	10	10	252570	309
1	36	36	490880	140
2	10	10	390030	1336
3	180	180	210265	108

E. COMPARISON WITH THE STATE-OF-THE-ART METHODS

To validate the effectiveness of our proposed hybrid methodology, we compared our optimal performance against recent state-of-the-art (SOTA) studies focusing on detection of healthcare fraud. Although our literature review identifies a range of methodologies, this quantitative comparison strictly isolates benchmarks that yield supervised predictive metrics (e.g., ROC-AUC, AUPRC) for direct mathematical evaluation. Direct comparisons are challenging due to variations in the datasets and class imbalance, but we evaluated our metrics against benchmark studies that used similar Medicare claims data [16]. The comparative analysis prioritizes the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and the Area Under the Precision-Recall Curve (AUPRC) over single-point metrics such as Precision, Recall, and the F1-score. F1-scores provide insight into model performance at an arbitrary classification threshold (typically 0.5). They do not capture an algorithm's predictive ability across diverse environments. In contrast, ROC-AUC and AUPRC are threshold-agnostic [17]. They are aggregate metrics that measure the total area under the curve across all possible thresholds from 0.0 to 1.0. As we deal with the severe class imbalance inherent in Medicare claims data, AUPRC is considered the critical evaluation metric in the study, since it is evaluated without accounting for True Negatives [17]. In addition, it provides a measure of the model's ability to accurately isolate the rare minority class (fraud) without skewing.

As shown in Table 5, the proposed SQL-based feature engineering with a class-weighted Regularized Logistic Regression achieves a highly competitive ROC-AUC of 0.94 and an AUPRC of 0.73. Our approach almost matches or outperforms the predictive capabilities of complex gradient

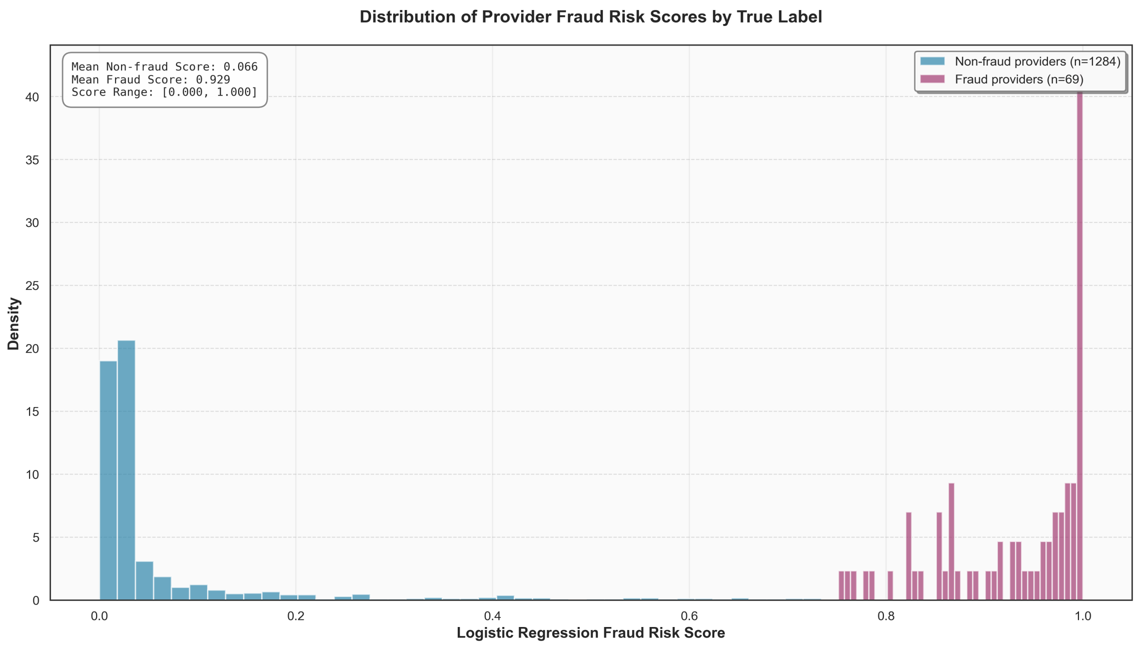


FIGURE 7: Histogram shows the distribution of logistic regression fraud risk scores for providers. Scores for non-fraudulent providers cluster near zero, while scores for fraudulent providers shift toward higher values.

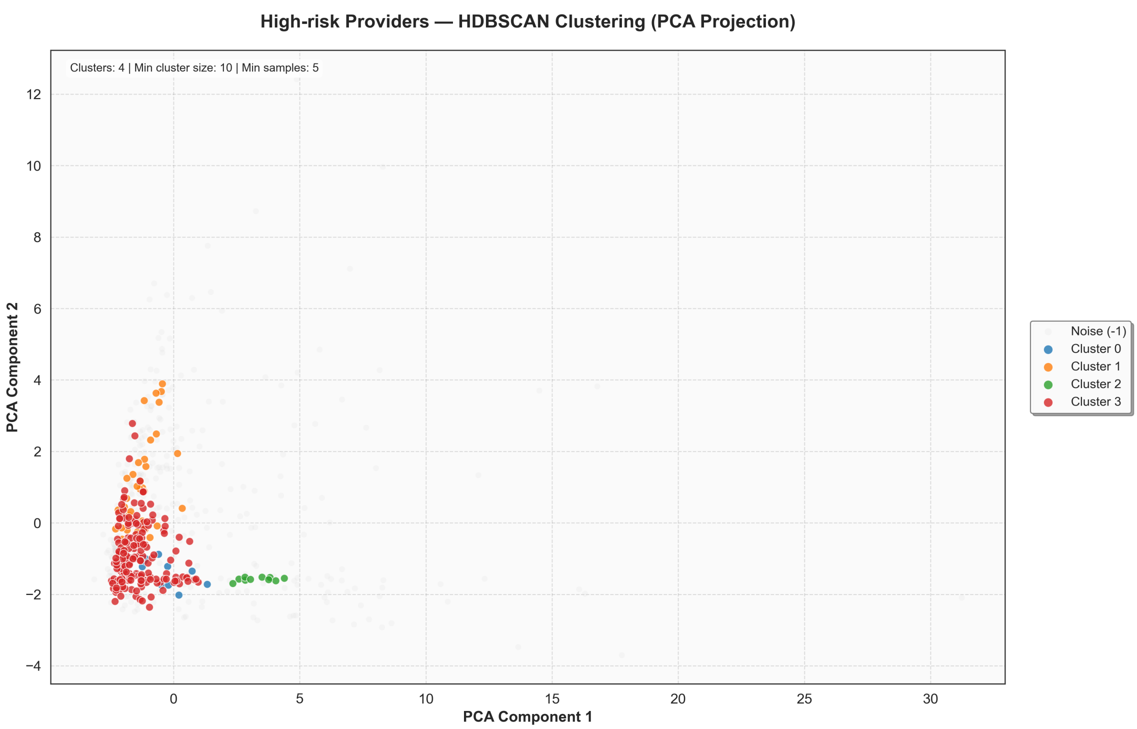


FIGURE 8: Two-dimensional PCA projection of standardized provider-level features for high-risk providers. Points are colored by HDBSCAN cluster label, with noise providers shown in grey.

TABLE 5: Performance Comparison with State-of-the-Art Fraud Detection Models

Study	Methodology	Dataset	Best ROC-AUC	Best AUPRC	Model Interpretability	Typology Mapping
Nabrawi & Alanazi [13]	Tree-based models & ANN	Saudi Claims	90.00%	Not Reported	Low (Black-Box)	No
Bounab et al. [16]	SMOTE-ENN + Tree Ensembles	Medicare Part B	99.00%	1.00	Low (Black-Box)	No
Hancock et al. [17]	GBDTs (Medicare Part D, Part B)	Medicare Claims	95.88%, 95.69%	0.80, 0.71	Low (Black-Box)	No
De Meulemeester et al. [18]	Latent Semantic Analysis Embeddings + Unsupervised + SHAP	Belgian Claims	82.00%	Not Reported	High (Explainable)	No
Kanksha et al. [19]	Isolation Forest	Medicare Claims	94.32%	Not Reported	Moderate	No
Proposed Approach	**SQL-Features + Reg. LR**	**Medicare (IP/OP)**	**94.00%**	**0.73**	**High (Linear / Rules)**	**Yes (HDBSCAN, PCA)**

boosting ensembles (GBDTs) proposed in recent literature. For example, our AUPRC of 0.73 outperforms the maximum AUPRC (0.71) reported by Hancock et al. [17] on Medicare Part B data, despite our use of a highly interpretable linear model. Furthermore, our results emphasize the necessity of Precision-Recall metrics over standard Accuracy. However, studies like Kanksha et al. [19] report accuracies exceeding 98%, such metrics often fail to capture minority-class detection in severely imbalanced fraud domains.

V. CONCLUSION AND FUTURE WORK

This study shows that a comprehensive and hybrid pipeline provides an effective framework to identify and characterize complex Medicare fraud. Our results pertain to classical machine learning methods, specifically regularized logistic regression, which outperforms complex tree-based models. Our strategy follows contemporary research that combines association rules, supervised models, and unsupervised clustering, rather than relying on a single model [1]- [4]. The regularized logistic regression model achieved an exceptional ROC-AUC of 0.94 and an AUPRC of 0.73, indicating that the prediction of fraudulent cases is driven more by feature engineering than by algorithm complexity [1], [2], [13]. Furthermore, our analysis goes beyond binary classification by successfully isolating distinct fraud typologies. Deep learning offers additional performance gains from a healthcare perspective, but requires greater computational power and resources [15]. This research offers a practical benchmark for real-world investigative systems by translating complex claims data into actionable groups such as high-revenue inpatient overcharging or high-volume outpatient overbilling [12], [20]. Ultimately, integrating a structured feature store with both supervised and unsupervised learning enables healthcare organizations to develop explainable, high-performance tools to combat sophisticated financial abuse, following best practices to restore integrity to public insurance programs [3], [4].

We have encountered a class imbalance problem, as most cases are labeled non-fraudulent, and our models are becoming biased toward normal cases. The complexity in feature engineering is profound. We tried to resolve the class imbalance problem using 50% of the fraudulent and 50% of the non-fraudulent cases from the train dataset. Also, we had computational limitations, so we did not integrate the deep learning models, as it would have cost a lot of time and power. The dataset we worked on covered only one year of patients' data, which might not be sufficient to obtain a holistic result. If we could work on the dataset in the expanded timeframe, the models could produce more mature output. We could not verify the flagged cases as fraudulent, as we need to verify that further with the Subject Matter Expert/Healthcare Professional. Currently, the approach is split test-train-based and, therefore, does not provide any real-time fraud scoring on patient data.

We can further enhance those models' performance by effectively tuning hyperparameters or by focusing on SQL-based feature engineering to improve performance metrics. Most studies have confirmed that SQL feature engineering is a crucial step that plays a significant role in improving evaluation scores [1], [2], [13]. We can also collaborate with a healthcare professional to assess the output of the model to flag fraudulent cases by validating the association rule mining on those claims. Subject-matter experts can review the top 100 claims or providers from fraud-type clusters and help refine the rule definitions. They can also help us ensure which clusters belong to which actionable schemes and provide more reliable labels. We can definitely test on the multi-year dataset, as it will provide more insights for fraud detection and can be integrated with deep learning models, despite being computationally intensive. Then we can move on to building a real-time fraud-scoring system. We will embed the current SQL feature store and the model training into the streaming architecture. New claims will automatically be added to the staging layer; all aggregates will be updated near real time. Based on the training set, the model will assign fraud scores to new claims and providers as they arrive. We can also integrate live dashboards where we can see the claims flagged as fraudulent for suspicious behavior and non-fraudulent cases as well. Integrating all of the above would lead to an innovative system for capturing atypical claims and providers in the healthcare industry, ensuring greater transparency.

DATA AVAILABILITY STATEMENT

The Medicare claims dataset analyzed during this study is publicly available and can be accessed through the Zenodo repository at https://doi.org/10.5281/zenodo.18138102 [35].

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